can do more harm than good. To build reliable ventilators, manufacturers would have to use components such as sensors, microprocessors and valves that were imported. Sourcing quality components during a time of global restrictions was one of the many hurdles for the manufacturers, that some tried to get around by developing some of the components indigenously.

Noccarc (then Nocca Robotics), an IIT Kanpur incubated startup managed to develop, prototype, test, and bring to the market an ICU ventilator within four months, helped by a crack team of mentors and using the same technologies that everyone used during the pandemic, Zoom and WhatsApp. The Startup Incubation and Innovation Centre (IISC) has on its board Professor Amitabha Bandyopadhyay and Srikant Sastri, authors of a book titled *The Ventilator Project* that traces step-by-step, all the obstacles and challenges overcome by the team to ship the product. The book has something for everyone including entrepreneurs, engineers, investors and cricket fans. The book is a roadmap of how to get things done in India, through global and local restrictions, in times of uncertainty. There are also insightful interviews which are like cameo appearances. Most importantly, there is a model for sustaining the accelerated development of world class products, and of reinventing silicon valley in India. The ball was set in motion when Professor Amitabha Bandyopadhyay sent out an SOS to all IIT Kanpur incubated startups on 16 March, looking for any projects that could help the nation address the pandemic. Noccarc responded initially with a proposal to manufacture ventilators, and then to design and develop their own ventilator. This was an ambitious and difficult project, but also the need of the hour. The team was testing and validating the ventilator within 90 days.

The first working prototype was actually built within two days, and demonstrated on 25 March. It was built using repurposed materials from the local markets in Pune, including repurposed drone parts. We had a chance to ask the founders of Noccarc, Nikhil Kurale and Harshit Rathore, to explain the components required for the initial prototype and how they were sourced: "The components used for the early prototype were diaphragm pumps, pneumatic fittings, electronic circuits, pressure sensors, touchscreen display, 3D printed plastic parts and sheet metal parts. Keeping in mind the immediate need to develop the prototype and the limitations imposed due to lockdown, these were sourced locally in Pune and from neighbouring cities."

Initially, Noccarc was looking to import their custom PCBs, but found suppliers in Pune who could meet their requirements, "the PCBs were manufactured by Epitome Components and assembled by Frontline Electronics Limited."

Right from the early days, Noccarc was helped on its quest by a group of Twitter users known as the Caring Indians. The group helped give visibility to the project in the critical first few days, and provided the initial networking to get the project going. Srikant Sastri and Amitabha Bandyopadhyay explain the role that the Caring Indians had to play in the early days of the project, "Caring Indians' is a crowdsourcing, social initiative that helped provide resources during the lockdown. This included masks and PPE kits for healthcare workers and others. This group was brought together to create a pool of product designs in collaboration with experts from academia and industry. It also helped bring together some of the country's best resources to develop cutting-edge healthcare solutions. Their requests were channelled through Twitter. Once we had connected with them, they helped us get through to some important people at an early stage, especially in the first 10 days of the project. They helped with the initial brainstorming, from the manufacturing readiness perspective, brought us information on other attempts that were being made across the globe, and connected us to some key government stakeholders. They also helped convert this project into a high profile one. Manish Mundra, a celebrated movie producer, talked about our project on a talk show with Smt. Smriti Irani. This, in turn, helped generate many

GEAR UP FOR THE PANDEMIC

Thermometers: Accuracy, speed of the readout, and ease of use are the main considerations. Replace the mercury-based one with an infrared device.

http://dgit.in/cvdt

Vaporisers: The ones with a face mask attachment are more comfortable to use. Capacity, time taken to mist and features are other things to look for.

http://dgit.in/cvdv

Pulse oximeters: Accuracy is the most important consideration. For the elderly, look for the devices with easy to read displays that are large, clear and legible.

http://dgit.in/cvdp

BP Monitors: Technically known as sphygmomanometers, the main thing to look out for here is ease of use, and of course accuracy. Pick from the models across the jump.

http://dgit.in/cvdb

Oxygen concentrators: While oxygen concentration machines are available online, do note that these machines should be used only under medical supervision.

http://dgit.in/cvdo